

HERMES: A Unified Self-Driving World Model for Simultaneous 3D Scene Understanding and Generation

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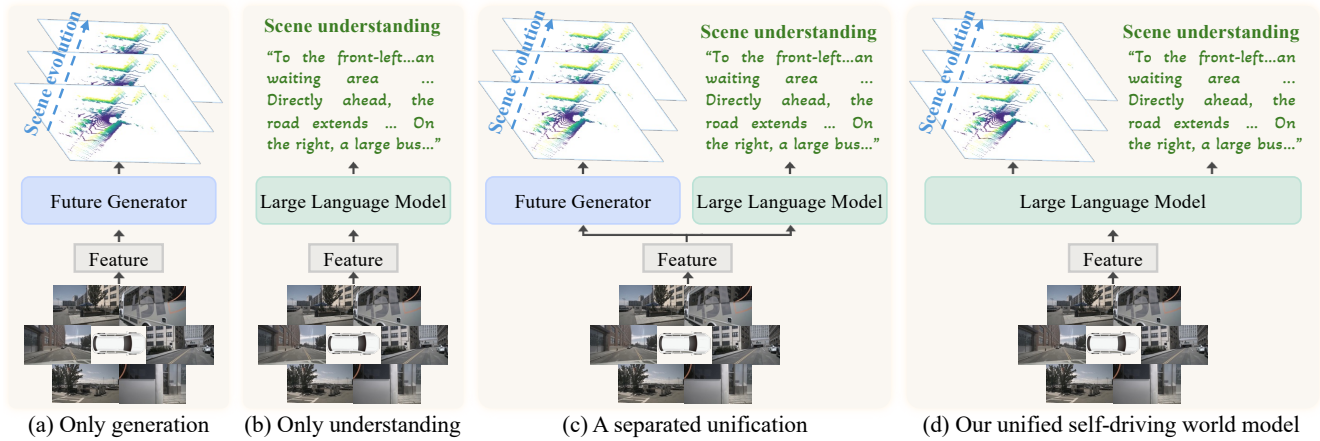


Figure 1. (a) Previous driving world models focus on generative scene evolution prediction. (b) Large language models for driving are limited to scene understanding. (c) A straightforward unification manner using the generator and large language models separately with a shared feature. (d) The proposed simple framework unifies 3D scene understanding and generates scene evolution based on given actions.

Abstract

Driving World Models (DWMs) have become essential for autonomous driving by enabling future scene prediction. However, existing DWMs are limited to scene generation and fail to incorporate scene understanding, which involves interpreting and reasoning about the driving environment. In this paper, we present a unified Driving World Model named **HERMES**. We seamlessly integrate 3D scene understanding and future scene evolution (generation) through a unified framework in driving scenarios. Specifically, HERMES leverages a Bird’s-Eye View (BEV) representation to consolidate multi-view spatial information while preserving geometric relationships and interactions. We also introduce world queries, which incorporate world knowledge into BEV features via causal attention in the Large Language Model, enabling contextual enrichment for understanding and generation tasks. We conduct comprehensive studies on nuScenes and OmniDrive-nuScenes datasets to validate the effectiveness of our method. HERMES achieves

state-of-the-art performance, reducing generation error by 32.4% and improving understanding metrics such as CIDEr by 8.0%. The model and code will be publicly released at <https://github.com/LMD0311/HERMES>.

1. Introduction

Driving World Models (DWMs) [12, 16, 51, 66] have become increasingly important in autonomous driving for their ability to predict future scene evolutions. These models simulate potential changes in the surrounding environment, enabling the vehicles to forecast risk, optimize routes, and make timely decisions in dynamic situations. Among the various modalities, point clouds [15, 23, 27, 61] naturally preserve the geometric relationships between different objects and their surroundings, making them well-suited for accurately describing scene evolutions [21, 54, 63, 72].

However, despite the progress in scene generation, a crucial limitation of current DWMs is their inability to incorporate scene understanding fully. Specifically, while these DWMs excel at predicting how the environment will evolve

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(Fig. 1(a)), they are hard to interpret and describe the environment, answer questions about it, or provide relevant contextual information (i.e., VQA, scene description).

Recently, vision-language models (VLMs) [6, 26, 29] have achieved remarkable advancements in general vision tasks by leveraging world knowledge and causal reasoning capabilities and have been successfully applied in autonomous driving scenes [41, 47]. As shown in Fig. 1(b), these driving VLMs are capable of performing tasks such as answering complex queries about the driving environment, generating descriptions of scenes, and reasoning about the relationships between various entities. However, while they improve understanding of the current driving environment, they still lack predictive capabilities for how the scene will evolve. This gap limits their effectiveness in autonomous driving, where both 3D scene understanding and future scene prediction are necessary for informed decision-making. This naturally gives rise to the question: *how can world knowledge and future scene evolutions be seamlessly integrated into a unified world model?*

Driven by the above motivation, in this paper, we propose a unified world model that connects both understanding and generation tasks. Our method is referred to as **HERMES**, as illustrated in Fig. 1(d), distinguishes itself from conventional methods, which typically specialize in either generation or scene understanding (e.g., VQA and caption). **HERMES** extends the capabilities of large language models (LLMs) to simultaneously predict future scenes and understand large-scale spatial environments, particularly those encountered in autonomous driving. However, constructing such a unified model is a highly non-trivial problem, as it requires overcoming several key challenges:

Large spatiality in multi-view. LLMs typically face max token length limits, especially in autonomous driving, where multiple surrounding views must be processed (e.g., six-view images in the nuScenes dataset [4]). Directly converting these multi-view images into tokens would exceed the token limit and fail to capture the interactions among different views. To address this, we propose to tokenize the input via a Bird’s-Eye View (BEV) representation. It offers two key benefits: 1) BEV effectively compresses the surrounding views into a unified latent space, thus overcoming the token length limitation while retaining key spatial information. 2) BEV preserves geometric spatial relationships between views, allowing the model to capture interactions between objects and agents across multiple perspectives.

The integration between understanding and generation. A straightforward way to unify scene understanding and generation would be to share the BEV features and apply separate models for understanding (via LLMs) and generation (via a future generator), as presented in Fig. 1(c). However, this approach fails to leverage the potential interactions between understanding and generation. Moreover,

the separate processing of these tasks hinders the optimization process, resulting in suboptimal performance. To address this, we propose to initialize a set of world queries using the raw BEV features (before LLM processing). These queries are then enhanced with world knowledge from text tokens through causal attention in the LLM. As a result, by using the world-knowledge-enhanced queries to interact with the LLM-processed BEV features via a *current to future link*, we ensure that the generated scene evolutions are enriched with world knowledge, effectively bridging the gap between generation and understanding.

By consolidating 3D scene understanding and future scene generation within a single framework, **HERMES** establishes a unified representation that seamlessly accommodates both tasks, offering a holistic perspective on driving environments. This marks a significant step toward a unified DWM, demonstrating the feasibility of integrated driving understanding and generation. Extensive experiments validate the effectiveness of our **HERMES** in terms of both tasks. Notably, our method significantly reduces the error by 32.4% compared to the current state-of-the-art (SOTA) method [61] for generation. Additionally, for the understanding task, our approach outperforms the SOTA [47] by 8.0% under the CIDEr metric on the challenging OmniDrive-nuScenes dataset [47].

Our major contributions can be summarized as follows:

1) In this paper, we propose **HERMES**, which tames the LLM to understand the autonomous driving scene and predict its evolutions simultaneously. To the best of our knowledge, this is the first world model that can unify the 3D understanding and generation task; 2) We introduce world queries to capture and integrate world knowledge from text tokens, ensuring that the generated scene evolutions are not only contextually aware but also enriched with world knowledge. This scheme effectively bridges the gap between the understanding and generation tasks, enabling a more coherent and accurate prediction of future scenes.

2. Related Work

World Models for Driving. Driving World Models (DWMs) [14] have gained considerable attention in autonomous driving for obtaining comprehensive environmental representation and predicting future states based on action sequences. Current research mainly focuses on the generation, whether in 2D [33, 50, 69] or 3D [34, 35].

Specifically, most pioneering 2D world models perform a video generation for driving scenarios. GAIA-1 [16] first introduced a learned simulator based on an autoregressive model. Recent work further leverages the large scale of data [20, 60, 64] and more powerful pre-training models, significantly enhancing generation quality regarding consistency [11, 51], resolution [12, 20], and controllability [24, 32, 53, 66]. Concurrently, some studies aim

to generate 3D spatial information for future scenes to provide geometric representations that can benefit autonomous driving systems. Occworld [68] focuses on future occupancy generation and ego planning using spatial-temporal transformers, which has been adapted to other paradigms including diffusion [13, 44], rendering [2, 18, 57], and autoregressive transformer [52]. Additionally, some approaches [21, 54, 63, 72] propose future point cloud forecasting as a world model, among which, ViDAR [61] using images to predict future point clouds through a self-supervised manner.

However, existing DWMs overlook the explicit understanding capacity of the driving environment. This paper aims to propose a unified world model that can both comprehend the scenario and generate scene evolution.

Large Language Models for Driving. Large Language Models (LLMs) exhibit impressive generalization and extensive world knowledge derived from vast data, showcasing remarkable capabilities across various tasks [9, 55, 65]. This has led researchers to explore their applications in autonomous driving, and current studies [8, 36, 40, 49] primarily focus on using LLMs to understand driving scenarios and make perceptual or decision-making outputs. For instance, DriveGPT4 [56] processes front-view video input to predict vehicle actions and provide justifications via an LLM. DriveLM [41] leverages LLMs for graph-based visual question-answering (VQA) and end-to-end driving. ELM [70] enhances the spatial perception and temporal modeling through space-aware pre-training and time-aware token selection. OmniDrive [47] introduces a benchmark with extensive VQA data labeled by GPT-4 [1] and utilizes Q-Former to integrate 2D pre-trained knowledge with 3D spatial. Despite significant advancements and the emergence of various language-based methods, the application of LLMs in driving remains mainly limited to understanding and text modeling. In this paper, we aim to tame the LLM to understand the autonomous driving scene and predict its future evolutions simultaneously.

3. Preliminaries

This section revisits the driving world models and the Bird’s Eye View representation as preliminary.

The Driving World Models (DWMs) seek to learn a general representation of the world from large-scale unlabeled driving data by forecasting future scenarios, enabling the model to grasp the data distribution of real situations. Specifically, given an observation $\mathcal{O}_t$ at time t , the model forecasts information about the next observation $\mathcal{O}_{t+1}$. The framework of DWMs can be summarized as follows:

$$\mathcal{L}_t = \mathcal{E}(\mathcal{O}_t), \mathcal{L}_{t+1} = \mathcal{M}(\mathcal{L}_t), \mathcal{O}_{t+1} = \mathcal{D}(\mathcal{L}_{t+1}), \quad (1)$$

where $\mathcal{E}$ and $\mathcal{D}$ represent the encoder and decoder for the scene, while the world predictor $\mathcal{M}$ maps the latent state

$\mathcal{L}_t$ to the next time step $\mathcal{L}_{t+1}$. Together, these components follow the workflow of $\mathcal{O}_t \rightarrow \mathcal{L}_t \rightarrow \mathcal{L}_{t+1} \rightarrow \mathcal{O}_{t+1}$.

Bird’s-Eye View (BEV) has emerged recently as a unified representation offering a natural candidate view. The BEVFormer series [25, 58] exemplifies this approach by leveraging cross-attention to improve 3D-2D view transformation modeling, resulting in robust BEV representations. This BEV feature maintains geometric spatial relationships between views, enabling the model to capture interactions among objects and agents from various perspectives. Additionally, BEV representations are ideal for integrating visual semantics and surrounding geometry, making them well-suited for understanding generation unification.

This paper focuses on unifying the 3D scene understanding and generation by using current multi-view as the observation to generate future point clouds, which inherently maintain accurate geometric relationships among objects and their environments within a BEV-based representation.

4. HERMES

This paper presents HERMES, a unified framework for driving scenarios understanding and generating. Our HERMES serving as a world model that predicts scenes’ point cloud evolution based on image observations and facilitates detailed scene comprehension. The pipeline of our method is illustrated in Fig. 2. We begin with multi-view input images I_t , which are encoded for semantic information using a BEV-based tokenizer and then processed by a Large Language Model (LLM). The LLM predicts the next token based on user instructions to interpret the current autonomous driving scenario. We integrate world queries into the sequence to transfer world knowledge from the conversations to the generation task. The *current to future link* generates future BEV features, and the shared render predicts the current scene point cloud P_t (as an auxiliary task) and generates Δt future scenes from P_{t+1} to $P_{t+\Delta t}$.

4.1. World Tokenizer and Render

The world tokenizer encodes the world observation, i.e., the current multi-view images, into a compressed continuous BEV representation, which is then further processed by the LLM. Conversely, the Render [45, 59, 71] converts BEV features into point clouds to generate geometric information about the scenario. Both modules are detailed as follows.

BEV-based World Tokenizer. To preserve geometric spatial relationships between views and rich semantic information for LLM inputs, we adopt a BEV-based world tokenizer $\mathcal{E}$. Specifically, the multi-view images I_t at time t are passed through a CLIP image encoder [7, 31] and a single frame BEVFormer v2 [58] without modification. The obtained BEV feature $\mathcal{F}_t^{bev} \in \mathbb{R}^{w \times h \times c}$ captures the semantic and geometric information, where w and h denote the scale of encoded scene, with larger values indicating greater de-

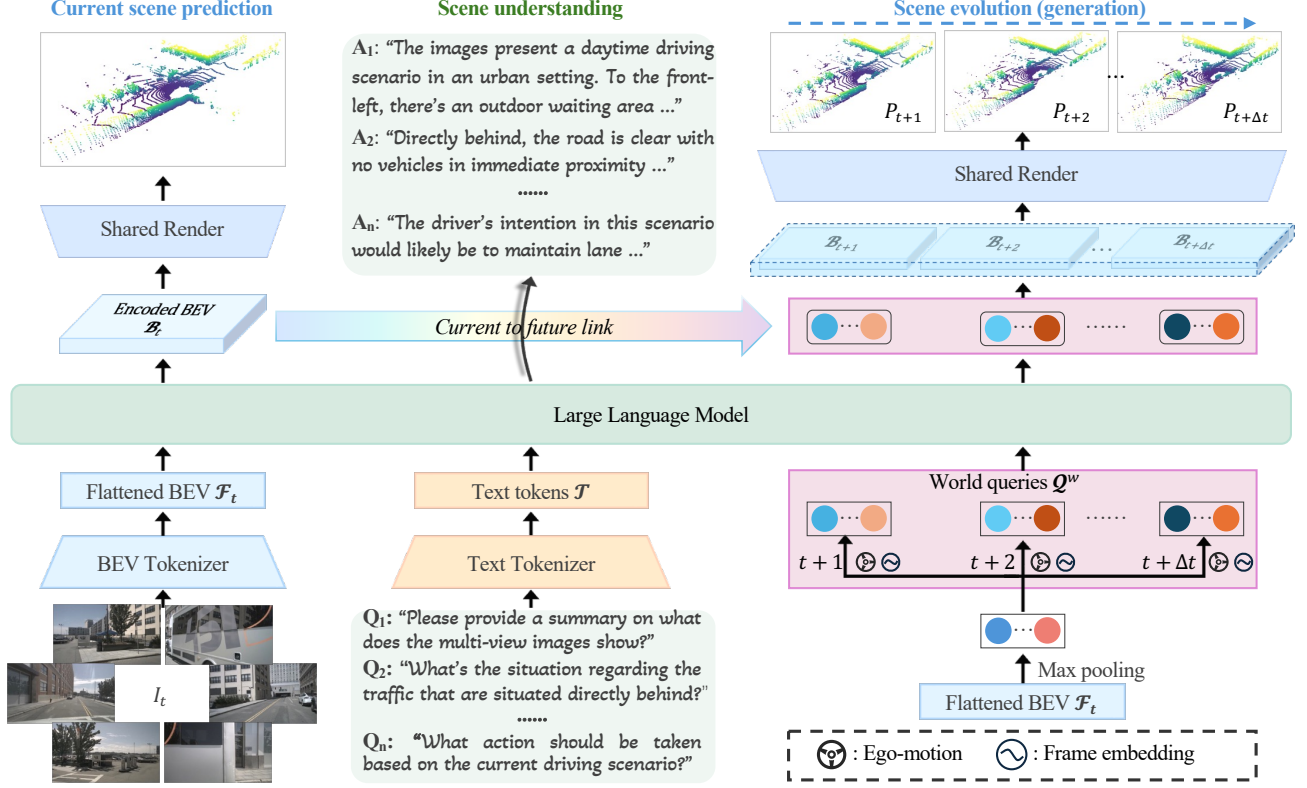


Figure 2. The pipeline of our HERMES. The BEV tokenizer converts multi-view I_t into flattened BEV $\mathcal{F}_t$, which are fed into the large language model (LLM). The LLM interprets user instructions $\mathcal{T}$ and generates textual responses by leveraging its understanding of driving scenes as world knowledge. A group of world queries $\mathcal{Q}^w$ are appended to the LLM input sequence. Encoded BEV $\mathcal{B}_t$ and world queries generate future BEV ($\mathcal{B}_{t+1}, \dots, \mathcal{B}_{t+\Delta t}$) via a *current to future link*, and the shared Render generates point clouds evolution.

tail, and c is the channel dimension of BEV. However, such a feature, often containing tens of thousands of tokens, is too large for the LLM [62]. To address this, we implement a down-sampling block that reduces $\mathcal{F}_t^{bev}$ by two times, resulting in a compressed shape of $\mathbb{R}^{\frac{w}{4} \times \frac{h}{4} \times (c \times 4)}$. When the LLM is required, the down-sampled feature will be flattened and projected to $\mathcal{F}_t \in \mathbb{R}^{L_{bev} \times C}$, where $L_{bev} = \frac{w}{4} \times \frac{h}{4}$.

BEV-to-Point Render. We introduce a simple BEV-to-Point Render $\mathcal{R}$, aiming to map the aforementioned down-sampled feature to the scene point cloud P_t . Specifically, we first up-sample the compressed BEV feature (or encoded BEV $\mathcal{B}_t \in \mathbb{R}^{L_{bev} \times (c \times 4)}$ after processing of the LLM and an out-projection) to the shape of $\mathbb{R}^{w \times h \times c}$ using nearest neighbor interpolation and convolutions. To address the absence of height information in the BEV feature, we reshape the input to $\mathbb{R}^{w \times h \times z \times \frac{c}{2}}$ by adding an extra height dimension. We then apply a series of 3D convolutions to reconstruct the volumetric feature $\mathcal{F}_t^{vol} \in \mathbb{R}^{w \times h \times z \times c'}$, where z is the height and c' is the output channel dimension. Finally, we construct rays $\{\mathbf{r}_k\}_{k=1}^K$ according to the LiDAR setup of the dataset and use differentiable volume rendering to compute the depth for each ray.

The rendering process models the environment as an im-

plicit signed distance function (SDF) field to capture intricate geometric details accurately [45, 59, 71]. Given a ray $\mathbf{r}_k$ originating from $\mathbf{o}$ and directed along $\mathbf{t}_k$, we discretize it into n sampled points $\{\mathbf{p}_i = \mathbf{o} + d_i \mathbf{t}_k \mid i = 1, \dots, n \text{ and } 0 \leq d_i < d_{i+1}\}$, where $\mathbf{p}_i$ corresponds to a location in 3D space, determined by its depth d_i along the ray. For each sampled point, we retrieve a local feature embedding $\mathbf{f}_i$ from the volumetric representation $\mathcal{F}_t^{vol}$ via trilinear interpolation. Subsequently, a shallow MLP ϕ_{SDF} is used to predict the SDF value $s_i = \phi_{\text{SDF}}(\mathbf{p}_i, \mathbf{f}_i)$. With the predicted SDF values, the rendered depth $\tilde{d}(\mathbf{r}_k)$ is computed through a weighted integration of all sampled depths by $\tilde{d}(\mathbf{r}_k) = \sum_{i=1}^n w_i d_i$, where $w_i = T_i \alpha_i$ [45] represents an unbiased, occlusion-aware weight. The transmittance $T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$ accumulates the survival probability of photons up to the j -th sample, $\alpha_i = \max(\frac{\sigma_t(s_i) - \sigma_t(s_{i+1})}{\sigma_t(s_i)}, 0)$ indicates the opacity, and $\sigma_t(x) = (1 + e^{-tx})^{-1}$ is a sigmoid modulated by a learnable parameter t .

4.2. Unification

This section introduces the unification of world understanding and future scene generation within our HERMES. The Large Language Model (LLM) interprets driving scenarios

from world tokenizer outputs ($\mathcal{F}_t$) based on user instructions. Δt groups of world queries gather knowledge from conversations, aiding in generating scene evolution.

Large Language Model. As shown in Fig. 2, the LLM is pivotal to our HERMES, modeling BEV inputs $\mathcal{F}_t$, parsing user instructions, acquiring world knowledge from real driving scenario inquiries, and generating predictions. We utilize the LLM within the widely used InternVL2 [5].

Understanding. Following prior work [29, 30], we project the flattened BEV to a shape of $\mathbb{R}^{L_{bev} \times C}$ and into the feature space of the LLM using a two-layer MLP, where L_{bev} is the input BEV length, and C is the channel dimension of LLM. For prompts on the current scene, we tokenize them into distinct vocabulary indices and text tokens $\mathcal{T}$ for processing by the LLM. Like existing multi-modal language models [5, 26], HERMES responds to user queries about the driving environment, providing scene descriptions and answers to visual questions. The LLM understands the scene through a next-token prediction approach.

Generation. Predicting and generating future changes based on observations of the current moment requires the model to have an exhaustive understanding of the world. To endow the LLM with future-generation capability, we propose a world query technique, which links world knowledge to future scenarios and improves information transfer between the LLM and the Render. As in Fig. 2, we outline the generation process in terms of LLM input and output.

For the input to the LLM, we utilize Δt groups of world queries $\mathcal{Q}^w \in \mathbb{R}^{(\Delta t \times n) \times C}$, where n is the number of queries per group and C represents the channel dimension of LLM. We emphasize the importance of proper feature initialization for effective learning. Thus, we employ a max pooling to derive the world queries from the peak of the BEV feature $\mathcal{F}_t$, yielding $\mathcal{Q} \in \mathbb{R}^{n \times (c \times 4)}$. The $\mathcal{Q}$ is then copied Δt times as query groups $\{\mathcal{Q}_i | i = 1, \dots, \Delta t\}$. To further enable controllable future generation, we encode the ego-motion condition to e_{t+i} , which describes the planned future positions and heading of the ego-vehicle from the current to the i -th frame into high-dimensional embeddings. The ego-motion information e_{t+i} is then added to the corresponding queries $\mathcal{Q}_i$. Additionally, a frame embedding $FE \in \mathbb{R}^{\Delta t \times (c \times 4)}$ is incorporated by the broadcast mechanism to denote the prediction frames for which group of world queries is responsible. The world queries $\mathcal{Q}^w$ and flattened BEV $\mathcal{F}^t$ share the language-space projection layer (i.e., MLP) to project from $c \times 4$ to C for the language model channel. The $\mathcal{Q}^w \in \mathbb{R}^{(\Delta t \times n) \times C}$ can be computed as:

$$\mathcal{Q}^w = \text{MLP}(\text{Concat}[\mathcal{Q}_i + e_{t+i} | i = \{1, \dots, \Delta t\}] + FE). \quad (2)$$

The LLM’s causal attention mechanism (i.e., the later token can access the earlier information) allows world queries to access world knowledge derived from the understanding process. After being processed by the LLM feed-forwardly,

the encoded BEV feature and world queries are projected by a shared two-layer MLP from the channel dimension of LLM C back to the channel of $c \times 4$. Note that each group of world queries contains only n queries, which provide a sparse view of the future world, complicating the reconstruction of the future scene by the world Render. To address this, we propose the *current to future link* module, which employs cross-attention layers to inject world knowledge for future BEV features. Specifically, the *current to future link* module contains 3 cross-attention blocks for generating future BEV features. Each cross-attention block includes a cross-attention layer that uses the encoded BEV $\mathcal{B}_t$ from the LLM output as the query, with world queries for each scene serving as the value and key. A self-attention layer and a feed-forward network further process spatial information. The encoded BEV ($\mathcal{B}_t$) and generated future BEV features ($\mathcal{B}_{t+1}, \dots, \mathcal{B}_{t+\Delta t}$) are sent to a shared world Render and obtain point cloud from P_t to $P_{t+\Delta t}$.

4.3. Training Objectives

To perform auto-regressive language modeling, we employ Next Token Prediction (NTP) to maximize the likelihood of text tokens, following the standard language objective:

$$\mathcal{L}_N = - \sum_{i=1} \log P(\mathcal{T}_i | \mathcal{F}_t, \mathcal{T}_1, \dots, \mathcal{T}_{i-1}; \Theta), \quad (3)$$

where $P(\cdot | \cdot)$ represents the conditional probability modeled by the weights Θ , $\mathcal{F}_t$ is the flattened BEV feature for the input frame, and $\mathcal{T}_i$ denotes the i -th text token.

For point cloud generation, we supervise the depths of various rays $d(\mathbf{r}_k)$ using only L1 loss:

$$\mathcal{L}_D = \sum_{i=0}^{\Delta t} \lambda_i \frac{1}{N_i} \sum_{k=0}^{N_i} |d(\mathbf{r}_k) - \tilde{d}(\mathbf{r}_k)|, \quad (4)$$

where λ_i is the loss weight for frame $t + i$, and N_i is the number of rays in the point clouds for frame $t + i$. The total loss for HERMES is given by $\mathcal{L} = \mathcal{L}_N + 10\mathcal{L}_D$.

5. Experiments

5.1. Dataset and Evaluation Metric

Datasets. 1) **NuScenes** [4] is a widely used autonomous driving dataset, which includes 700 training scenes, 150 validation scenes, and 150 test scenes. We use six images and the point clouds captured by surrounding cameras and the LiDAR, respectively. 2) **NuInteract** [67] is a newly proposed language-based driving dataset with dense captions for each image and scene. With $\sim 1.5\text{M}$ annotations, it supports various tasks, such as 2D perception and 3D visual grounding. 3) **OmniDrive-nuScenes** [47] supplements nuScenes with high-quality caption and visual question-answering (QA) text pairs generated by GPT4. Considering

Table 1. The comparison of our HERMES and understanding/generation specialist models. L/C/T refers to LiDAR/camera/text, respectively. We report MTETOR, CIDEr, and ROUGE for understanding tasks, and Chamfer distance for 0-3s on the (OmniDrive-)nuScenes validation set, following ViDAR [61]. [†] denotes results from ViDAR, while scores for GPT-4o and LLaVA-OV are sourced from DriveMM [17].

Method	Reference	# LLM Params	Modality	Generation				Understanding		
				0s ↓	1s ↓	2s ↓	3s ↓	MTETOR ↑	ROUGE ↑	CIDEr ↑
Only Generation										
4D-Occ [†] [21]	CVPR 23	-	L→L	-	1.13	1.53	2.11	Unsupported		
ViDAR [61]	CVPR 24	-	C→L	-	1.12	1.38	1.73			
Only Understanding										
GPT-4o [19]	-	-	C→T					-	0.223	0.244
LLaVA-OV [22]	arXiv 24	7B	C→T					-	0.221	0.284
OmniDrive [47]	CVPR 25	7B	C→T			Unsupported		0.380	0.326	0.686
OmniDrive-2D [47]	CVPR 25	7B	C→T					0.383	0.325	0.671
OmniDrive-BEV [47]	CVPR 25	7B	C→T					0.356	0.278	0.595
Unified Understanding and Generation										
Separated unification	-	1.8B	C→T&L	0.60	0.84	1.08	1.37	0.384	0.327	0.745
HERMES (ours)	-	1.8B	C→T&L	0.59	0.78	0.95	1.17	0.384	0.327	0.741

the high quality and rich VQA annotations, we perform the understanding training and evaluation on the OmniDrive-nuScenes description and conversation data.

Evaluation Metric. For understanding tasks, we utilize widely used METEOR [3], CIDEr [43], and ROUGE [28] metrics to compute similarities between generated and ground-truth answers at the word level. For generation evaluation, we follow previous work [61] and use Chamfer Distance to measure precision in generated point clouds, considering only points within the range of [-51.2m, 51.2m] on the X- and Y-axes, and [-3m, 5m] on the Z-axis.

5.2. Main Results

We compare HERMES with understanding [19, 22, 47] and generation [21, 61] specialist models in Tab. 1, which demonstrates competitive performance on both tasks and promote strong unification.

For future point cloud generation, both 4D-Occ and ViDAR utilize a 3s history horizon, while our HERMES only relies on the current frame, achieving significant improvements. Remarkably, with only multi-view inputs, HERMES is capable of generating a more accurate representation of the scene geometry for predicting future evolution, resulting in $\sim 32\%$ Chamfer Distance reduction in 3s point clouds compared to ViDAR. It should be noted that ViDAR utilizes a carefully designed latent render and an FCOS3D [48] pre-trained backbone, while HERMES uses simple volumetric representation. Furthermore, HERMES can simultaneously understand the current scenario, which is a crucial capability for driving systems but is challenging for existing driving world models.

For 3D scene understanding, we continuously achieve highly competitive results in caption quality compared to understanding specialists. For example, we notably outperform OmniDrive by 8% on the CIDEr metric and excel in

MTETOR and ROUGE. Note that OmniDrive leverages extensive 2D pre-training data [10], supervision from 3D objects [46], and lane detection. The OmniDrive-BEV uses LSS [38] to transform perspective features into a BEV feature and SOLOFusion [37] for temporal modeling. While it utilizes a BEV representation, its scene understanding is limited, likely due to insufficient data for BEV-based image-text alignment. Further compared to the Separated unification model, our unified approach yields significantly better generation results while maintaining strong understanding performance. This demonstrates successful cross-task knowledge transfer and consistent scene modeling within our compact, unified HERMES.

5.3. Ablation Study

Unless otherwise specified, we perform ablation studies trained on a quarter of the nuScenes training scenes. Default settings are marked in green.

Analysis on understanding and generation interaction. Our HERMES achieves a strong and seamless unification of understanding and generation in driving scenes. We first analyze the relationship between these two processes. As shown in Tab. 2, we conduct experiments with four approaches: solely understanding, solely generation, a separated unification, and our method. The separated unification involves sharing the flattened BEV while using separate models for understanding and generation, as in Fig. 1(c). We find that HERMES achieves highly competitive results compared to training on one task, with a minor performance gap (e.g., 0.002 difference on MTETOR/ROUGE and a 0.03 chamfer distance gap on the 3s generation). Nevertheless, our approach shows better results at 0-1s, indicating ongoing optimization challenges in the unified understanding and generation stage. Ultimately, our HERMES outperforms the separated unification in generation results, as the lat-

Table 2. Ablation on interaction of tasks.

Under.	Gen.	Generation				Understanding		
		0s ↓	1s ↓	2s ↓	3s ↓	MTETOR ↑	ROUGE ↑	CIDEr ↑
✓	-	-	-	-	-	0.379	0.323	0.728
-	✓	0.651	0.988	1.313	1.687	-	-	-
Separated unify		0.663	1.095	1.476	1.875	0.377	0.321	0.722
✓	✓	0.645	0.984	1.333	1.718	0.377	0.321	0.720

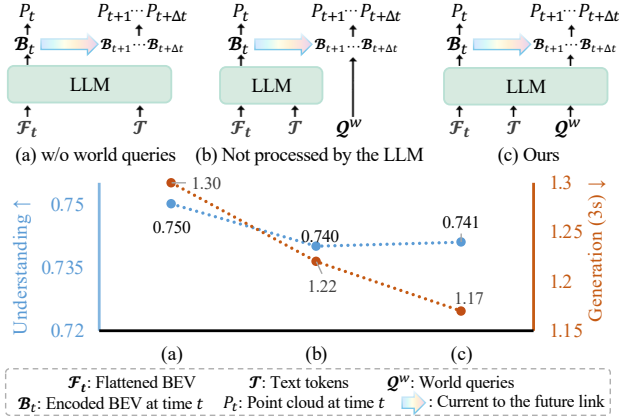


Figure 3. The effect of world queries for understanding (CIDEr) and generation (3s chamfer distance) is trained on full data.

ter fails to exploit the potential interactions between understanding and generation, and the separation hinders optimization and leads to suboptimal performance.

Analysis on the effect of the world queries. We then validate the efficacy of world queries (Q^w), as shown in Fig. 3. Without world queries (Fig. 3(a)), the sequences $B_{t+1}, \dots, B_{t+\Delta t}$ are generated by directly integrating future ego motion into B_t . It can be found that introducing world queries significantly improves future generation capabilities, reducing the Chamfer Distance for 3s point cloud prediction by 10%. Although there is a slight 1% decrease in understanding performance (CIDEr), we attribute this to the increased optimization complexity from adding new informational parameters to the LLM. A comparison of Fig. 3(b)-(c) further reveals that Q^w processed through the LLM enhances generative performance, supporting their role in effective world knowledge transfer.

Analysis on generation length. In Tab. 3, we discuss the number of generated frames for future scenes. As the number increases, the generation results exhibit a slight drop, which we attribute to an optimization dilemma within the LLM. This suggests that a more efficient interaction method could be explored in the future. Additionally, we evaluate the auxiliary role of current point cloud prediction for future generation, as shown in the last two rows of Tab. 3. Predicting the current frame regularizes the encoded BEV (B_t) from the LLM outputs, enhancing future generation results. More importantly, training for current point cloud

Table 3. Ablation on generation length.

Second	0	1	2	3	Generation				Understanding		
					0s ↓	1s ↓	2s ↓	3s ↓	MTETOR ↑	ROUGE ↑	CIDEr ↑
✓	✓	-	-	-	0.607	0.944	-	-	0.379	0.323	0.725
✓	✓	✓	-	-	0.632	0.951	1.313	-	0.378	0.321	0.714
-	✓	✓	✓	-	1.078	1.397	1.779	-	0.378	0.321	0.717
✓	✓	✓	✓	✓	0.645	0.984	1.333	1.718	0.377	0.321	0.720

Table 4. Ablation on the source of world queries.

Pooling	Generation				Understanding		
	0s ↓	1s ↓	2s ↓	3s ↓	MTETOR ↑	ROUGE ↑	CIDEr ↑
Attn.	0.656	1.001	1.344	1.748	0.377	0.321	0.712
Avg.	0.660	0.996	1.348	1.741	0.376	0.321	0.715
Max	0.645	0.984	1.333	1.718	0.377	0.321	0.720

Table 5. Ablation on size for the flattened BEV.

BEV size	Generation				Understanding		
	0s ↓	1s ↓	2s ↓	3s ↓	MTETOR ↑	ROUGE ↑	CIDEr ↑
25 × 25	0.720	1.040	1.347	1.698	0.367	0.311	0.671
50 × 50	0.645	0.984	1.333	1.718	0.377	0.321	0.720

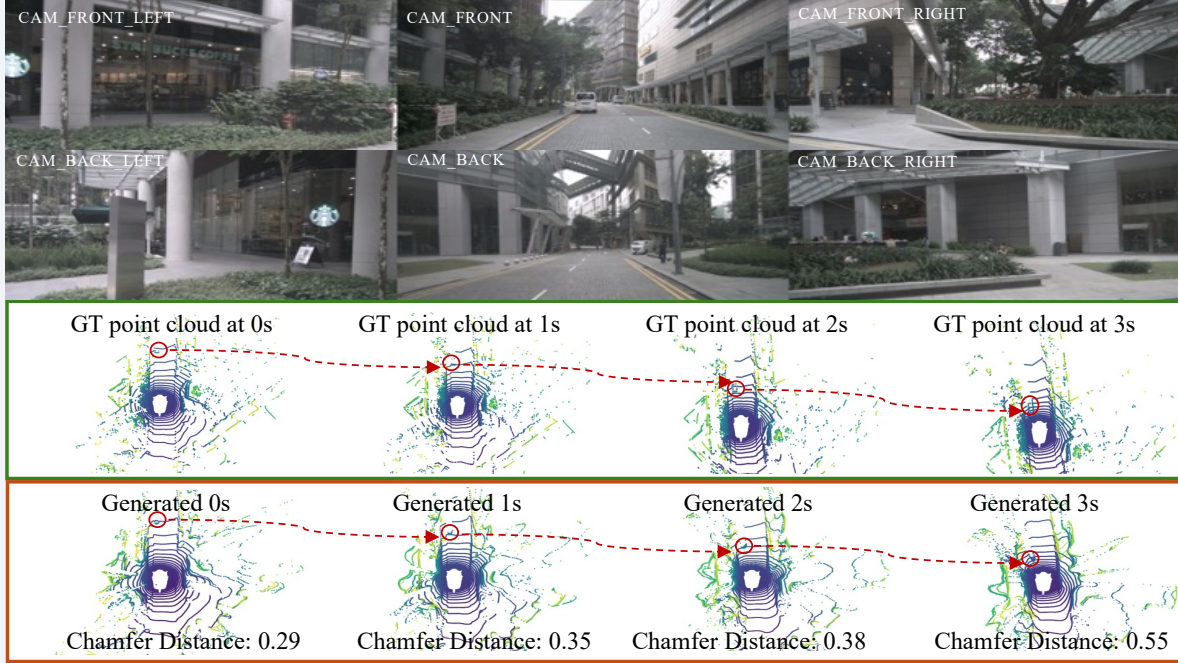
prediction does not add extra inference burden for future generations, serving as a practical auxiliary task.

Analysis on the source of world queries. We also assess the initialization of world queries in Tab. 4, including attention pooling [39, 42], adaptive average pooling and max pooling. It shows that world queries derived from the adaptive max pool of the flattened BEV F_t perform better in chamfer distance on 3s. We argue that the max pool effectively captures peak responses from F_t , whereas average or attention pooling may overly emphasize global information, potentially affected by background noise.

Analysis on the size of flattened BEV. We finally analyze the impact of flattened BEV (F_t) size by varying the BEV tokenizer’s downsampling multiplier, resulting in different feature lengths. Experiments with 8× and 4× downsampling yield F_t with spatial resolutions of 25 and 50, respectively. Tab. 5 demonstrates that a spatial size of 50 × 50 significantly improves CIDEr and 0s generation by 7.3% and 10% compared to 25 × 25. We attribute this improvement to reduced information loss compared to excessive downsampling, enhancing text comprehension and facilitating point cloud recovery/prediction. While further increasing the size of F_t might improve performance, we chose 50 × 50 as a trade-off due to LLM processing length limitations.

5.4. Qualitative Results

This section presents qualitative results on future generations and scene understanding, as illustrated in Fig. 4. Our



Q: Can you provide a summary of the current driving scenario based on the input images?

A: The vehicle is currently navigating through an urban environment, seemingly a **business district** with modern architecture and office buildings. **On the left**, the view is dominated by a **Starbucks Coffee** shop with a large **glass façade** and a **lush greenery barrier**, while **on the right**, an open plaza with more green spaces and **seated areas** suggests a pedestrian-friendly zone. The road ahead is clear, marked with well-defined lane lines, and there are no visible traffic signals or signs indicating any immediate need to stop or yield. The weather appears overcast, providing soft lighting without the glare of direct sunlight. Looking behind, the street is equally clear, with **no vehicles or pedestrians in close proximity**, indicating a low traffic condition.

Q: Is there a traffic signal in sight that the vehicle should be aware of?

A: **No, there is no traffic signal** visible in the front or rear views that the vehicle should be aware of at this moment.

Q: Are there any pedestrians close to the vehicle's path?

A: **No, there are no pedestrians close to the vehicle's path** on the sidewalks or crossing the streets.

Q: Can you see any vehicles in your rearview that you should be aware of?

A: The rear view is **clear of any close following vehicles**, allowing for a comfortable driving distance.

Q: Is there any oncoming traffic that might affect your driving?

A: There is **no oncoming traffic visible** that would affect the current driving conditions.

Q: What action should be taken in the current driving scenario?

A: Given the clear road ahead and the current driving conditions, the vehicle should maintain a moderate speed and stay in the current lane. \n- Continue to monitor the surroundings for any unexpected pedestrians that may enter the roadway from the right side where the pedestrian plaza is located.\n- Regularly check the rearview and side mirrors to stay aware of any vehicles that may approach from behind, although the current rear view shows no immediate traffic.\n- Be prepared to react to any traffic signals or signs that may come into view, even though none are currently visible.

Figure 4. Qualitative results for future generation and scene understanding. From top to bottom, we display the multi-view input of the current scene, the ground truth scene evolution, the generated scene evolution, and the scene understanding result.

HERMES effectively captures future scene evolution, such as the van keeping close to the ego vehicle, noted in the red circle. Furthermore, HERMES exhibits a strong understanding of BEV inputs, as indicated by the green text, accurately identifying objects such as “Starbucks”.

6. Conclusion

This paper introduces **HERMES**, a simple yet effective unified Driving World Model that integrates 3D scene understanding and future scene generation within a single framework. We effectively bridge the gap between understanding and generation by leveraging a Bird’s-Eye View representation and incorporating world queries enhanced through

large language models. Extensive experiments validate the effectiveness of the proposed HERMES, demonstrating significant improvements in future scene prediction accuracy and understanding metrics. We believe this work is a significant step toward a stronger unified Driving World Model.

Limitation. Although HERMES achieves promising results in unifying 3D scene understanding and generation, there are some limitations: 1) We have not explored perception tasks for autonomous driving within our framework. 2) Future image is also an important generation modality but is still under exploration. We left these in our future work.

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